

Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Computational physicist with a PhD in Physics and ten years of experience building and running simulation codes for relativistic plasmas. Author of Paramo, an open-source Fortran library that solves the Fokker-Planck equation coupled to radiative transfer. Led an upgrade of the PatchworkMHD code on Frontera and used gradient-based optimization to calibrate model parameters against observations. Nine publications and a NASA grant won and led as a postdoc.

Skills

Simulation: Kinetic plasma modeling (Fokker-Planck), radiative processes (synchrotron, inverse Compton), radiative transfer, relativistic MHD, hydrodynamics (SPH), Lagrangian particle dynamics, numerical PDEs, finite difference/volume methods, numerical linear algebra, parameter estimation, uncertainty quantification

HPC & Parallel: OpenMP, OpenACC (GPU offloading), MPI, SLURM, PBS, HDF5

ML & Data: Scientific machine learning, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Languages: C/C++, Python, Fortran 95, Rust, R, Java, SQL, Julia, Shell

Tools: Git, Docker, Jenkins, QGIS, Paraview, VisIt

Experience

Independent Contractor, Mercor.

Jan 2026 — Present

- Reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.

Spatial Data Scientist, TealWaters.

May 2025 — Jan 2026

- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.

Software Engineer, Paychex.

Apr 2022 — Jan 2024

- Developed and tested production data-pipeline software (Java, Python, SQL) with CI/CD.

Independent Researcher & Open-Source Developer, Self-directed.

Jan 2024 — Present

- Co-developed Tleco, a Rust/Python toolkit that solves the Fokker-Planck equation coupled to radiative transfer for relativistic outflows (ApJ 2024).
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library for wind-driven spore dispersal over hilly forest terrain (RcppParallel, OpenMP).

Postdoctoral Research Associate, Rochester Institute of Technology.

Feb 2021 — Apr 2022

- Led a 7-person team in an NSF-sponsored upgrade of PatchworkMHD, a C code for MHD simulations of supermassive black hole binaries, adding black hole spin with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo parameters against Fermi-LAT observations using gradient-based optimization, comparing simulation output with data and updating the model automatically; supervised a graduate student on the same problem.
- Collaborated on a multi-institutional HPC project on binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow, Purdue University.

Oct 2018 – Nov 2020

- Designed and wrote Paramo, an open-source Fortran 95 library that solves the Fokker-Planck equation for relativistic particle distributions coupled to radiative transfer. parallelized and optimized with OpenMP; GPU offload tested with OpenACC.
- Implemented implicit PDE solvers, numerical integration, and arbitrary-shape particle distributions without slowing the simulation.
- Wrote Python tools to analyze Paramo output and run parameter optimization; other groups have adopted Paramo, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals, one funded (\$68K) to model blazar radiation; mentored 3 graduate students and published 2 peer-reviewed papers.

Postdoctoral Research Fellow, Universidad Michoacana de San Nicolás de Hidalgo.

Jan – Sep 2018

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Graduate Research Assistant, Universitat de València.

Oct 2011 – Jul 2017

- Studied how magnetization of shocked plasma shapes the radiation spectra of relativistic internal shocks, using the C-SPEV simulation code.

Education

Universitat de València, *Ph.D. in Physics.*, Excellent Cum Laude..

Valencia, Spain

THESIS: *Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows*

Jul. 2017

Universidad Michoacana de San Nicolás de Hidalgo, *M.Sc. in Physics.*

Morelia, Mexico

THESIS: *Study of TOV stars with the SPH method*

Sep. 2011

Projects

Blazar Population ML, github.com/altjerue/ilwikak-ml.

Jan 2026 — Present

Scientific ML (Python, scikit-learn, PyTorch)

- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength observational data.
- Building a neural network to learn the mapping from physical jet parameters to observable SEDs accurately enough to compare with Tleco results.
- Forecasting blazar gamma-ray flares using a general-purpose time series foundation model, fine-tuned with Tleco-generated synthetic light curves.

WindsOfChange, github.com/UW-ForestMycobiomeLab/WindSporesHills.

Aug 2024 — Present

Open-Source Lagrangian Particle Simulation Library (R/C++)

- Developed C++ library implementing Lagrangian particle dispersion, atmospheric turbulence, and fluid flow over complex hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004).
- Parallelized particle tracking using RcppParallel and OpenMP; designed for scalability to tens of thousands of particles across kilometer-scale domains.

Tleco, github.com/zkdavis/Tleco.

Jan 2024 — Sep 2024

Open-Source Physics Simulation Library

- Multiscale/multiphysics simulation toolkit combining Rust and Python interfaces for modeling relativistic particle distributions via Fokker-Planck equations coupled with radiative transfer.
- Implements numerical PDE solvers, particle distribution algorithms, and spectral calculation methods for relativistic plasma physics.

Paramo, github.com/altjerue/paramo.

Oct 2018 — Apr 2022

Open-Source Physics Simulation Code (CONAHCYT Funded)

- Modular physics simulation library (Fortran 95 + OpenMP) modeling relativistic particle radiation; predecessor to Tleco.

Selected Publications (9 total, 3 first-author)

- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, **ApJ** **976**, 182, (2024), [arXiv:2405.17581](https://arxiv.org/abs/2405.17581).
- Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, **MNRAS** **513**, 5766–5779, (2022), [arXiv:2201.07790](https://arxiv.org/abs/2201.07790).
- **Rueda-Becerril, J. M.**, Harrison, A. O. & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, **MNRAS** **501**, 4092–4102, (2021), [arXiv:2009.02273](https://arxiv.org/abs/2009.02273).
- **Rueda-Becerril, J. M.**, Mimica, P. & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, **MNRAS** **468**, 1169–1182, (2017), [arXiv:1612.06383](https://arxiv.org/abs/1612.06383).